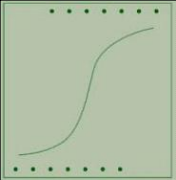




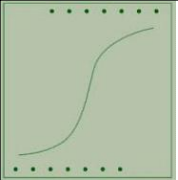
DESCRIBING CONTINGENCY TABLES





IxJ Contingency Table

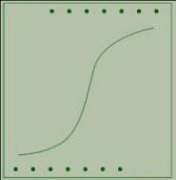
- Let X and Y denote two categorical response variables, X with I categories and Y with J categories.
- Classifications of subjects on both variables have IJ possible combinations.
- The cells of the table represent the IJ possible outcomes
- When the cells contain frequency counts of outcomes for a sample, the table is called a contingency table
- Another name is cross-classification table. A contingency table with I rows and J columns is called an $I \times J$ or I -by- J table.



Study. aspirin use and heart attacks

- ▶ 5 year randomized, blind study
- ▶ Two arms,
 - Placebo
 - Aspirin
- ▶ Arm1 – placebo: of 11,034, 18 had fatal heart attacks
- ▶ Arm2 – aspirin: of 11,037, 5 had it

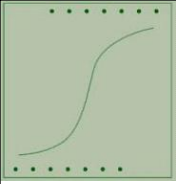
	Myocardial Infarction		
	Fatal Attack	Nonfatal Attack	No Attack
Placebo	18	171	10,845
Aspirin	5	99	10,933



- Joint distribution:
- Let π_{ij} $i=1, \dots, I; j=1, \dots, J$
 - the probability that (X, Y) occurs in the cell in row i and column j .
- marginal distributions: the row and column totals that result from summing the joint probabilities.
 - π_{i+} $i=1, \dots, I$ for the row variable
 - π_{+j} $j=1, \dots, J$ for the column variablewhere

$$\pi_{i+} = \sum_j \pi_{ij} \quad \text{and} \quad \pi_{+j} = \sum_i \pi_{ij}.$$

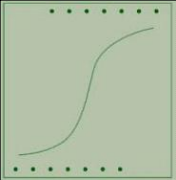
$$\sum_i \pi_{i+} = \sum_j \pi_{+j} = \sum_i \sum_j \pi_{ij} = 1.0.$$



Conditional probability

- one variable, say Y , is a response variable and the other X is an explanatory variable.
- for a fixed category of X , Y has a probability distribution.
- Given that a subject is classified in row i of X , $\pi_{j|i}$ denotes the probability of classification in column j of Y , $j=1, \dots, J$.
- The probabilities $\{\pi_{1|i}, \dots, \pi_{J|i}\}$ form the conditional distribution of Y at category i of X where

$$\sum_j \pi_{j|i} = 1.$$



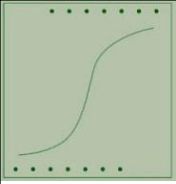
Independence

- Recall in probability:
 - two events A and B are independent if and only if $P(AB) = P(A) \cdot P(B)$
 - Two random variables are independent if and only if $f(x, y) = f(x) \cdot f(y)$
- Two categorical response variables are defined to be independent if all joint probabilities equal the product of their marginal probabilities,

$$\pi_{ij} = \pi_{i+} \pi_{+j} \quad \text{for } i = 1, \dots, I \quad \text{and} \quad j = 1, \dots, J. \quad (2.1)$$

- Thus, when X and Y are independent,

$$\pi_{j|i} = \pi_{ij} / \pi_{i+} = (\pi_{i+} \pi_{+j}) / \pi_{i+} = \pi_{+j} \quad \text{for } i = 1, \dots, I.$$



Independence

- Each conditional distribution of Y is identical to the marginal distribution of Y .
- Or for any $j=1, \dots, I$,

$$\pi_{j|1} = \dots = \pi_{j|I}$$
- Independence is then often referred to as **homogeneity** of the conditional distributions.

Population
parameters

π

π_{ij}

$\pi_{j|i}$

Sample

p

$p_{ij} = n_{ij}/n$

$p_{j|i} = p_{ij}/p_{i+} = n_{ij}/n_{i+}$

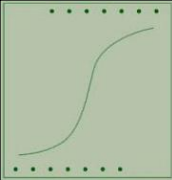
where

$$n = \sum_i \sum_j n_{ij}$$

$$n_{i+} = \sum_j n_{ij}$$

Table of Treatment by outcome

Table of Treatment by outcome				
Treatment	outcome			Total
Frequency				
Percent				
Row Pct		NonFatal		
Col Pct	Fatal Attack	Attack	No Attack	
Aspirin	n_{11} π_{11} $\pi_{1 1}$	n_{12} π_{12} $\pi_{2 1}$	n_{13} π_{13} $\pi_{3 1}$	n_{1+} π_{1+}
Placebo	n_{21} π_{21} $\pi_{1 2}$	n_{22} π_{22} $\pi_{2 2}$	n_{23} π_{23} $\pi_{3 2}$	n_{2+} π_{2+}
Total	n_{+1} π_{+1}	n_{+2} π_{+2}	n_{+3} π_{+3}	n_{++}



2.1.2 Sensitivity and Specificity

TABLE 2.2 Estimated Conditional Distributions for Breast Cancer Diagnoses

Breast Cancer	Diagnosis of Test		Total
	Positive	Negative	
Yes	0.82	0.18	1.0
No	0.01	0.99	1.0

Source: Data from W. Lawrence et al., *J. Natl. Cancer Inst.* 90: 1792–1800 (1998).

- Given that the subject has the disease, the conditional probability that the diagnostic test is positive is called the sensitivity;
- given that the subject does not have the disease, the conditional probability that the test is negative is called the specificity

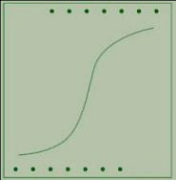
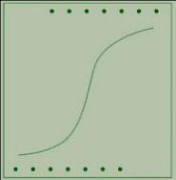


TABLE 2.2 Estimated Conditional Distributions for Breast Cancer Diagnoses

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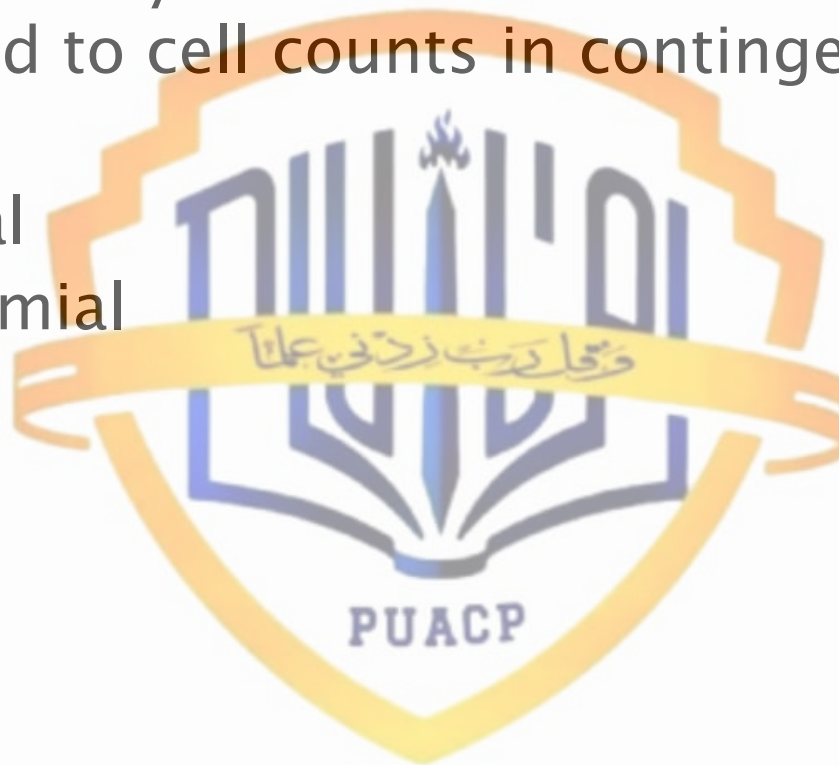
Source: Data from W. Lawrence et al., *J. Natl. Cancer Inst.* 90: 1792–1800 (1998).

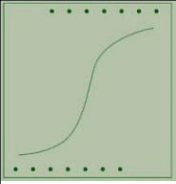
- the estimated sensitivity of combined mammography and clinical examination is 0.82.
 - Of women with breast cancer, 82% are diagnosed correctly.
- the estimated specificity is 0.99.
 - Of women not having breast cancer, 99% were diagnosed correctly.



2.1.4 Poisson, Binomial, and Multinomial Sampling

- ▶ The probability distributions introduced in Section 1.2 extend to cell counts in contingency tables.
 - Poisson
 - Binomial
 - Multinomial

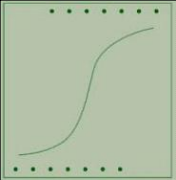




Poisson Sampling

- a Poisson sampling model treats cell counts $\{Y_{ij}\}$ as independent Poisson random variables with parameters $\{\mu_{ij}\}$.
- The joint probability mass function for $P(\bar{Y}_{ij} = n_{ij})$ potential outcomes $\{n_{ij}\}$ is then the product of the Poisson probabilities for the IJ cells, or

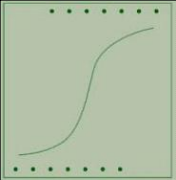
$$\prod_i \prod_j \exp(-\mu_{ij}) \mu_{ij}^{n_{ij}} / n_{ij}! .$$



Multinomial Sampling

- ▶ When the total sample size n is fixed but the row and column totals are not, a multinomial sampling model applies.
- ▶ The IJ cells are the possible outcomes.
- ▶ The probability mass function of the cell counts has the multinomial form

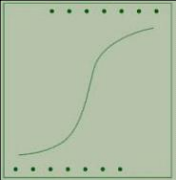
$$\left[\frac{n!}{n_{11}! \cdots n_{IJ}!} \right] \prod_i \prod_j \pi_{ij}^{n_{ij}}.$$



product multinomial sampling

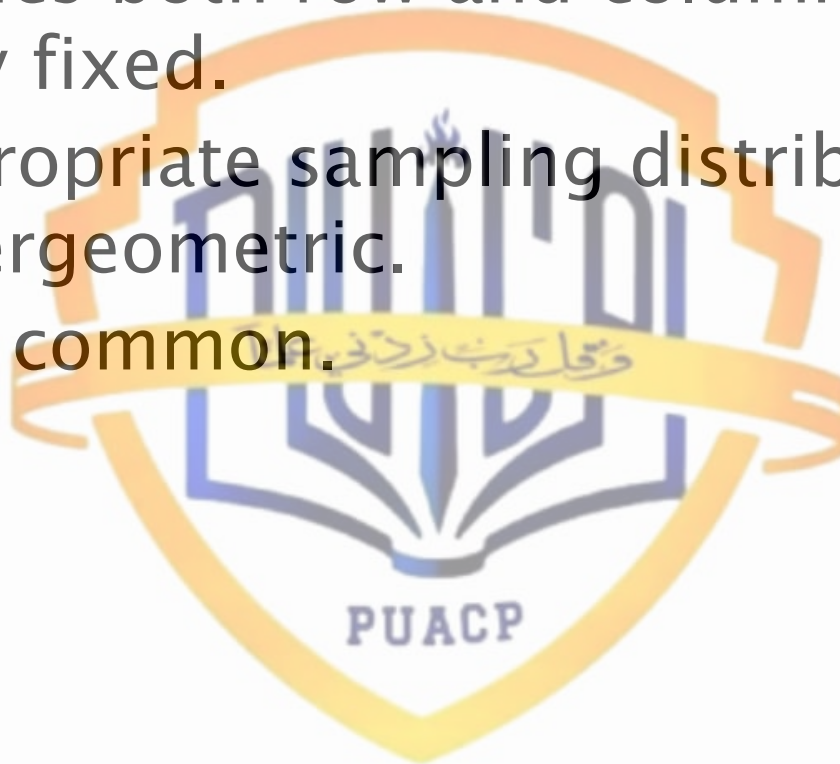
- Observations on a response Y occur separately at each setting of an explanatory variable X .
- This case normally treats row totals as fixed, and for simplicity, we use the notation $n_i = n_{i+}$
- Suppose that the n_i observations on Y at setting i of X are independent, each with probability distribution
- The counts $\{n_{ij}, j = 1, \dots, J\}$ satisfying $\sum_j n_{ij} = n_i$ then have the multinomial form $\{\pi_{1|i}, \dots, \pi_{J|i}\}$
- When samples at different settings of X are independent, the joint probability function for the entire data set is the product of the multinomial functions

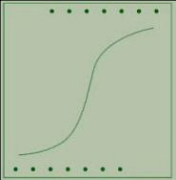
$$\frac{n_i!}{\prod_j n_{ij}!} \prod_j \pi_{j|i}^{n_{ij}}.$$



Hypergeometric distribution

- ▶ Sometimes both row and column margins are naturally fixed.
- ▶ The appropriate sampling distribution is then the hypergeometric.
- ▶ It is less common.



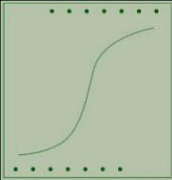


2.1.5 Seat Belt Example

- ▶ Researchers in the Massachusetts Highway Department plan to study
 - the relationship between seat-belt use (yes, no) and outcome of an automobile crash (fatality, nonfatality) for drivers involved in accidents on the Massachusetts Turnpike.
- ▶ Data is summarized as

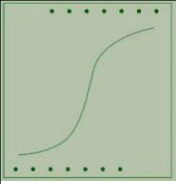
TABLE 2.4 Seat-Belt Use and Results of Automobile Crashes

Seat-Belt Use	Result of Crash	
	Fatality	Nonfatality
Yes		
No		



Seat Belt Example

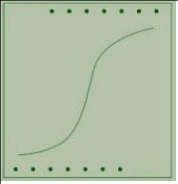
- Design 1: They plan to catalog all accidents on the turnpike for the next year, classifying each according to these variables.
 - > The total sample size is then a random variable
 - > Treat the numbers of observations at the four combinations as independent Poisson $\{\mu_{11}, \mu_{12}, \mu_{21}, \mu_{22}\}$
- Design 2: researchers randomly sample 200 police records of crashes on the turnpike in the past year and classify each according to seat-belt use and outcome of crash.
 - > $n=200$ is fixed
 - > Multinomial ($n, \{\pi_{11}, \pi_{12}, \pi_{21}, \pi_{22}\}$)



Seat Belt Example

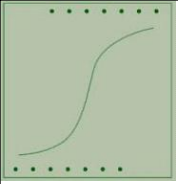
- ◉ Design 3: The researchers might instead randomly sample 100 records of accidents with a fatality and randomly sample 100 records of accidents with no fatality.
 - This approach fixes the column totals in Table 2.4 at 100.
 - They might then regard each column of Table 2.4 as an independent binomial sample. (retrospective)
- ◉ Design 4: (traditional experimental design)
 - takes 200 subjects and randomly assigns 100 of them to wear seat belts; the 200 then all are forced to have an accident. (prospective)
 - The recorded results would then be independent binomial samples in each row, with fixed row totals of 100 each.

This May be unethical for humans.



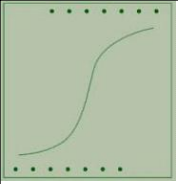
2.1.6 Types of Study

- ◉ **Observational studies**
 - Cohort study
 - Prospective cohort
 - Retrospective cohort
 - Case-control study
 - Cross-sectional study
- ◉ **Treatment studies (experimental studies, prospective)**
 - Randomized controlled trial
 - Nonrandomized trial (quasi-experiment)
 - Clinical trials



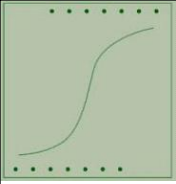
prospective cohort study

- ◉ A **prospective cohort study** is a research effort that
 - follows over time groups of individuals
 - who are similar in some respects (e.g., all are working adults)
 - but differ on certain other characteristics (e.g., some smoke and others do not)
 - and compares them for a particular outcome (e.g., lung cancer)
- ◉ It can be more expensive than a case-control study.



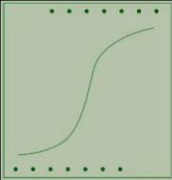
retrospective cohort study

- ◉ A retrospective cohort study, also called a historic cohort study, is a medical research study in which
 - > the medical records of groups of individuals who are alike in many ways but differ by a certain characteristic (for example, female nurses who smoke and those who do not smoke) are compared for a particular outcome (such as lung cancer).
- ◉ As is obvious, Retrospective Cohort has the benefits of being cheaper and less time consuming.
- ◉ The resources are mainly directed at collection of data only.



Case-control study

- ◉ **Case-control** is a type of epidemiological study design.
 - > Case-control studies are used to identify factors that may contribute to a medical condition
 - > by comparing subjects who have that condition (the 'cases') with patients who do not have the condition (the 'controls')
 - > but are otherwise similar.
- ◉ Case-control studies are a relatively inexpensive and frequently-used type of epidemiological study that can be carried out by small teams



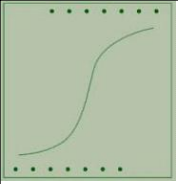
A case control study

- ▶ In 20 hospitals in London, England, patients admitted with lung cancer in the preceding year were queried about their smoking behavior.
- ▶ For each of the 709 patients admitted, researchers studied the smoking behavior of a noncancer patient at the same hospital of the same gender and within the same 5-year grouping on age.

TABLE 2.5 Cross-Classification of Smoking by Lung Cancer

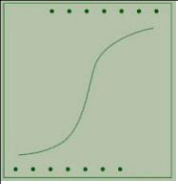
Smoker	Lung Cancer	
	Cases	Controls
Yes	688	650
No	21	59
Total	709	709

Source: Based on data reported in Table IV, R. Doll and A. B. Hill, *British Med. J.*, Sept. 30, 1950, pp. 739–748.



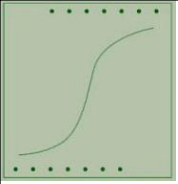
lung cancer with smoking behavior

- distribution of lung cancer is fixed by the sampling design, and the outcome measured is whether the subject ever was a smoker.
- It is a retrospective design to “look into the past”
- Often, the two samples are matched, as in this study.
- Sometimes the samples of cases and controls are independent rather than matched.
- For those in Table 2.5 with lung cancer, the proportion who were smokers was $688/709=0.970$, while it was $650/709=0.917$ for the controls.
- we cannot estimate the probability of lung cancer at each category of smoking behavior, without knowing the proportion of the population having lung cancer.



Cross-sectional study

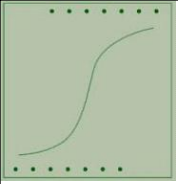
- ◎ Cross-sectional form a class of research methods that
 - > involve observation of some subset of a population of items all at the same time,
- ◎ The fundamental difference between cross-sectional and longitudinal studies is that cross-sectional studies take place at a single point in time and that a longitudinal study involves a series of measurements taken over a period of time. Both are a type of observational study.



Clinical trial

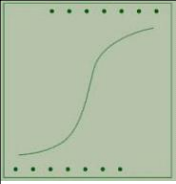
- ▶ A randomized controlled trial (RCT) is a type of scientific experiment most commonly used in testing the efficacy or effectiveness of healthcare services (such as medicine or nursing) or health technologies (such as pharmaceuticals, medical devices or surgery).
- ▶ This trial randomly allocates the subject to the respective groups e.g., smokers or non smokers

PUACP



2.2 COMPARING TWO PROPORTIONS

- Binary response such as (success, failure) for outcome of a medical treatment are common
- With two groups, a 2x2 contingency table displays the results
- We set the rows as groups and the columns are the categories of Y .
- We need to compare the groups:
 - > Difference of proportions
 - > Odds ratio
 - > Relative risk

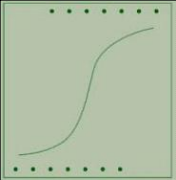


2.2.1 Difference of Proportions

- For subjects in row i , $\pi_{1|i}$ is the probability that the response has outcome in category 1 (“success”).
- With only two possible outcomes, $\pi_{2|i} = 1 - \pi_{1|i}$
- we use the simpler notation π_i for $\pi_{1|i}$.
- The difference of proportions of successes, $\pi_1 - \pi_2$, is a basic comparison of the two rows.
- Comparison on failures is equivalent to comparison on successes, since

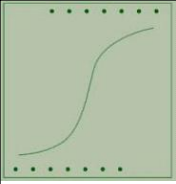
$$(1 - \pi_1) - (1 - \pi_2) = \pi_2 - \pi_1.$$

- $-1 \leq \pi_1 - \pi_2 \leq 1$



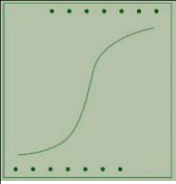
Difference of Proportions

- $\pi_1 - \pi_2 = 0$ when the rows have identical conditional distributions.
- When both variables are responses, conditional distributions apply in either direction.
- Example 1: a study comparing two treatments on the proportion of subjects who die,
 - > $\pi_1 - \pi_2 = 0.010 - 0.001 = 0.009$
- Example 2: another study comparing two treatments on the proportion of subjects who die, $\pi_1 = 0.410$, $\pi_2 = 0.401$.
 - > $\pi_1 - \pi_2 = 0.410 - 0.401 = 0.009$



2.2.2 Relative Risk

- A value $\pi_1 - \pi_2$ of fixed size may have greater importance when both π_i are close to 0 or 1 than when they are not.
- The ratio of proportions is also informative.
- The relative risk is defined to be the ratio π_1 / π_2
- It can be any nonnegative real number.
- A relative risk of 1.0 corresponds to independence.
- Example 1: the relative risks are $0.010 / 0.001 = 10.0$
- Example 2: $0.410 / 0.401 = 1.02$



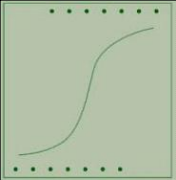
2.2.3 Odds Ratio

- For a probability π of success, the odds are defined to be

$$\Omega = \pi / (1 - \pi).$$

$$\pi = \Omega / (\Omega + 1).$$

- The odds are nonnegative, with $\Omega > 1.0$ when a success is more likely than a failure.
- Example: $\pi = 0.75$, $\Omega = 0.75 / 0.25 = 3.0$; a success is three times as likely as a failure, and we expect about three successes for every one failure.



Odds Ratio

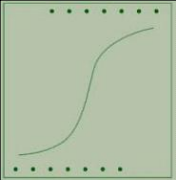
- Refer again to a 2x2 table. Within row i , the odds of success are $\Omega_i = \pi_i / (1 - \pi_i)$. The ratio
- of the odds Ω_1 and Ω_2 in the two rows,

$$\theta = \frac{\Omega_1}{\Omega_2} = \frac{\pi_1 / (1 - \pi_1)}{\pi_2 / (1 - \pi_2)}$$

- Or

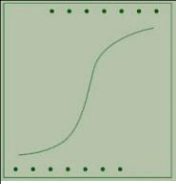
$$\theta = \frac{\pi_{11} / \pi_{12}}{\pi_{21} / \pi_{22}} = \frac{\pi_{11} \pi_{22}}{\pi_{12} \pi_{21}}.$$

where $\{\pi_{ij}\}$ are cell probabilities of joint distributions.



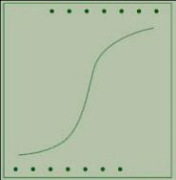
2.2.4 Properties of the Odds Ratio

- The odds ratio can equal any nonnegative number.
- independence of X and Y $\Leftrightarrow \Omega_1 = \Omega_2 \Leftrightarrow \text{OR } \theta = 1$
- When $\theta > 1$, subjects in row 1 are more likely to have a success than are subjects in row 2 $\Leftrightarrow \pi_1 > \pi_2$
- For instance, when $\text{OR } \theta = 4$, the odds of success in row 1 are four times the odds in row 2. This does not mean that the probability $\pi_1 = 4\pi_2$; that is the interpretation of a relative risk of 4.0.
- When $0 < \theta < 1$, $\pi_1 < \pi_2$
- When one cell has zero probability, θ equals 0 or ∞ .



OR

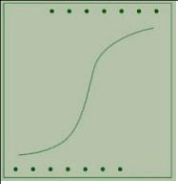
- Values of θ farther from 1.0 in a given direction represent stronger association.
- Two values represent the same association, but in opposite directions, when one is the inverse of the other.
- For instance, when $\theta=0.25$, the odds of success in row 1 are 0.25 times the odds in row 2, or equivalently, the odds of success in row 2 are $1/0.25=4.0$ times the odds in row 1.
- The 0.25 and 4 represents the same amount of associations, but in different directions.
- $\text{Log}(\theta)$ can be better to display the amount of association, for example
 $\log(0.25)=\log(1/4)=-\log(4)=-1.39$
 $\log(4)=1.39$



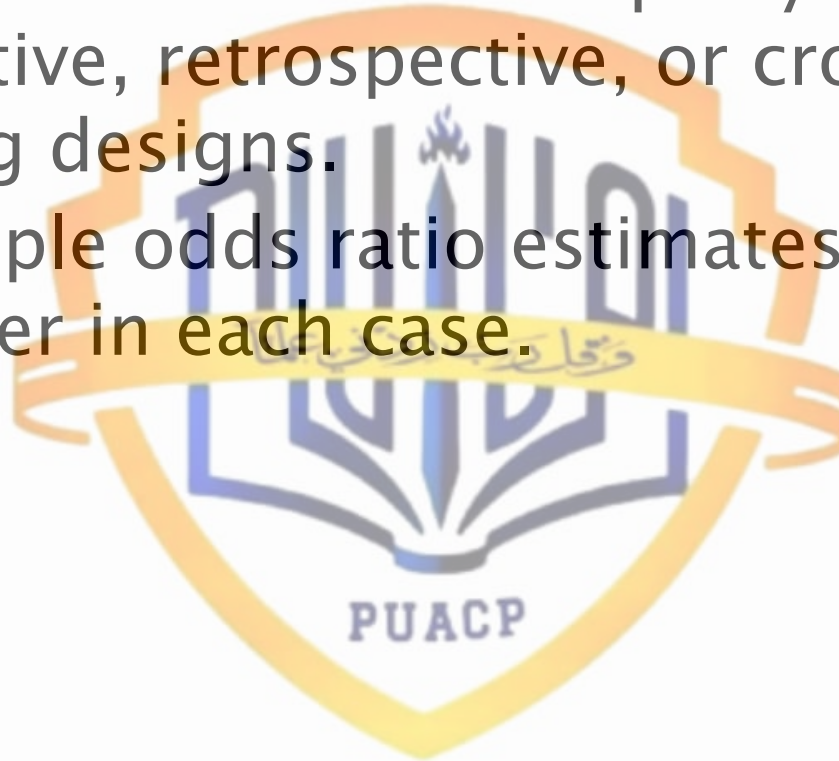
OR property

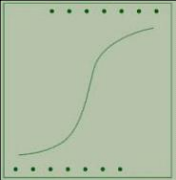
- ▶ The odds ratio does not change value when the orientation of the table reverses so that the rows become the columns and the columns become the rows.
- ▶ It is unnecessary to identify one classification as the response variable in order to use OR.
- ▶ In fact, although in terms of odds using $\pi_i = P(Y = 1 | X = i)$ one could just as well define it using reverse conditional probabilities.

$$\begin{aligned}\theta &= \frac{\pi_{11}\pi_{22}}{\pi_{12}\pi_{21}} = \frac{P(Y = 1 | X = 1)/P(Y = 2 | X = 1)}{P(Y = 1 | X = 2)/P(Y = 2 | X = 2)} \\ &= \frac{P(X = 1 | Y = 1)/P(X = 2 | Y = 1)}{P(X = 1 | Y = 2)/P(X = 2 | Y = 2)}.\end{aligned}\tag{2.6}$$



- ▶ In fact, the odds ratio is equally valid for prospective, retrospective, or cross-sectional sampling designs.
- ▶ The sample odds ratio estimates the same parameter in each case.





Sample odds ratio

$$\hat{\theta} = n_{11}n_{22}/n_{12}n_{21}.$$

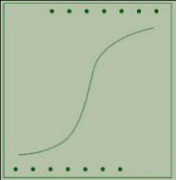
- ◉ It does not change
 - > when both cell counts within any row are multiplied by a nonzero constant
 - > or when both cell counts within any column are multiplied by a nonzero constant.
- ◉ An implication is that the sample odds ratio estimates the same characteristic OR, even when the sample is disproportionately large or small from marginal categories of a variable.

2.2.5 Aspirin and Heart Attacks

TABLE 2.1 Cross-Classification of Aspirin Use and Myocardial Infarction

	Myocardial Infarction		
	Fatal Attack	Nonfatal Attack	No Attack
Placebo	189	104	10,845
Aspirin	104	99	10,933

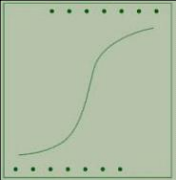
- $\pi_1 = 189/11034 = 0.0171$; $\pi_2 = 104/11037 = 0.0094$
- $\pi_1 - \pi_2 = 0.0077$; $\pi_1 / \pi_2 = 1.82$
- The proportion suffering heart attacks of those taking placebo was 1.82 times the proportion suffering heart attacks of those taking aspirin.
- $OR = 189 \cdot 10933 / (10845 \cdot 104) = 1.83$
- The odds of heart attack for those taking placebo was 1.83 times the odds for those taking aspirin.



2.2.6 Case-Control Studies and the Odds Ratio

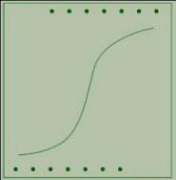
Smoker	Lung Cancer	
	Cases	Controls
Yes	688	650
No	21	59
Total	709	709

- With retrospective sampling designs, such as case control studies, it is possible to estimate conditional probabilities of form $P(X = i | Y = j)$
- Not possible $P(Y = j | X = i)$
- Impossible to estimate diff. risk or RR
- However, as OR is symmetric, we can get the estimate of OR



Smoker	Lung Cancer	
	Cases	Controls
Yes	688	650
No	21	59
Total	709	709

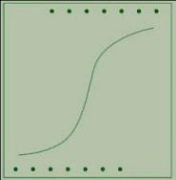
- the probability a subject was a smoker, given the subject had lung cancer = $688/709$
- the probability a subject was a smoker, given the subject did not have lung cancer = $650/709$
- the probability of lung cancer, given whether one smoked?
- we cannot estimate differences or ratios of probabilities of lung cancer.
- However, $OR = \frac{(688/709)/(21/709)}{(650/709)/(59/709)} = \frac{688 \times 59}{650 \times 21} = 3.0$.
- The estimated odds of lung cancer for smokers were 3.0 times the estimated odds for nonsmokers.



2.2.7 Relationship between Odds Ratio and Relative Risk

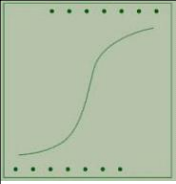
$$\text{odds ratio} = \text{relative risk} \left(\frac{1 - \pi_2}{1 - \pi_1} \right).$$

- ▶ Their magnitudes are similar whenever the probability π_i of the outcome of interest is close to zero for both groups.
- ▶ In this case, the odds ratio provides a rough indication of the relative risk when it is not directly estimable, such as in case-control studies.
- ▶ For example, if the probability of lung cancer is small regardless of smoking behavior, 3.0 is also a rough estimate of the relative risk;
 - smokers had about 3.0 times the relative frequency of lung cancer as nonsmokers.



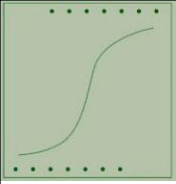
2.3 PARTIAL ASSOCIATION IN STRATIFIED 2x2 TABLES

- An important part of most studies, especially observational studies, is the choice of control variables.
- In studying the effect of X on Y, one should control any covariate that can influence that relationship.
- This involves using some mechanism to hold the covariate constant. Otherwise, an observed effect of X on Y may actually reflect effects of that covariate on both X and Y.
- The relationship between X and Y then shows confounding.
- Experimental studies can remove effects of confounding covariates by randomly assigning subjects to different levels of X, but this is not possible with observational studies.



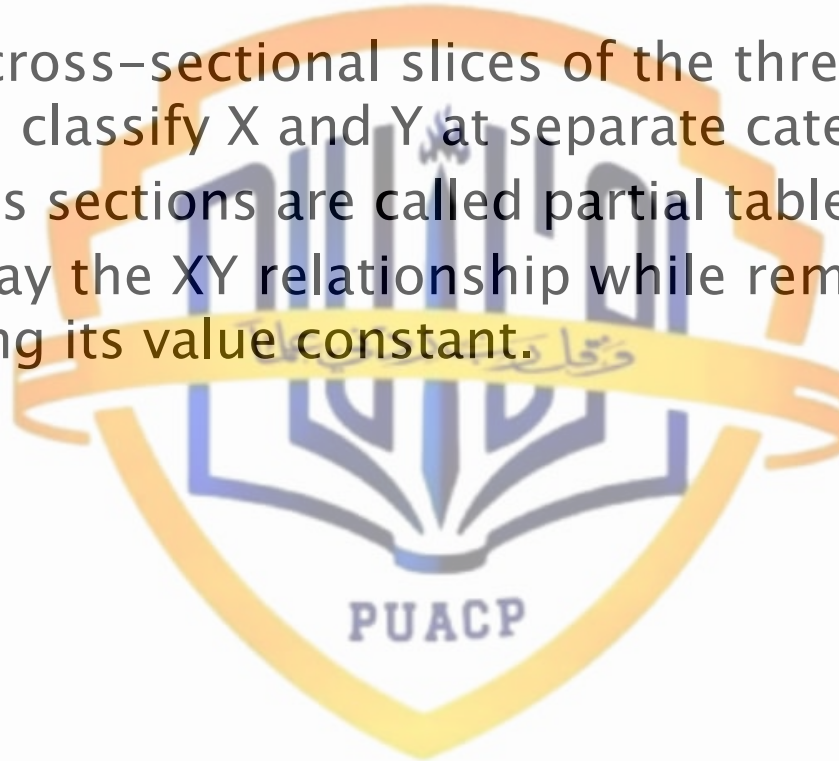
Confounding example

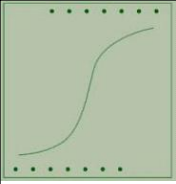
- ▶ Study: effects of passive smoking on lung cancer
- ▶ A cross-sectional study might compare lung cancer rates between nonsmokers whose spouses smoke and nonsmokers whose spouses do not smoke.
- ▶ The study should attempt to control for age, socioeconomic status, or other factors that might relate both to spouse smoking and to developing lung cancer.
- ▶ Otherwise, results will have limited usefulness.
- ▶ Spouses of nonsmokers may tend to be younger than spouses of smokers, and younger people are less likely to have lung cancer.
- ▶ Then a lower proportion of lung cancer cases among spouses of nonsmokers may merely reflect their lower average age.



2.3.1 Partial Tables

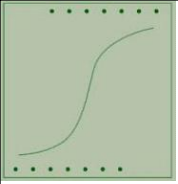
- ▶ We control for Z by studying the XY relationship at fixed levels of Z.
- ▶ Two-way cross-sectional slices of the three-way contingency table cross classify X and Y at separate categories of Z.
- ▶ These cross sections are called partial tables.
- ▶ They display the XY relationship while removing the effect of Z by holding its value constant.





marginal table

- ▶ The two-way contingency table obtained by combining the partial tables is called the XY marginal table.
- ▶ Each cell count in the marginal table is a sum of counts from the same location in the partial tables.
- ▶ The marginal table, rather than controlling Z, ignores it.
- ▶ The marginal table contains no information about Z.
- ▶ It is simply a two-way table relating X and Y but may reflect the effects of Z on X and Y.



conditional associations and marginal associations

- ▶ The associations in partial tables are called conditional associations, because they refer to the effect of X on Y conditional on fixing Z at some level.
- ▶ Conditional associations in partial tables can be quite different from associations in marginal tables.
- ▶ In fact, it can be misleading to analyze only marginal tables of a multiway contingency table.

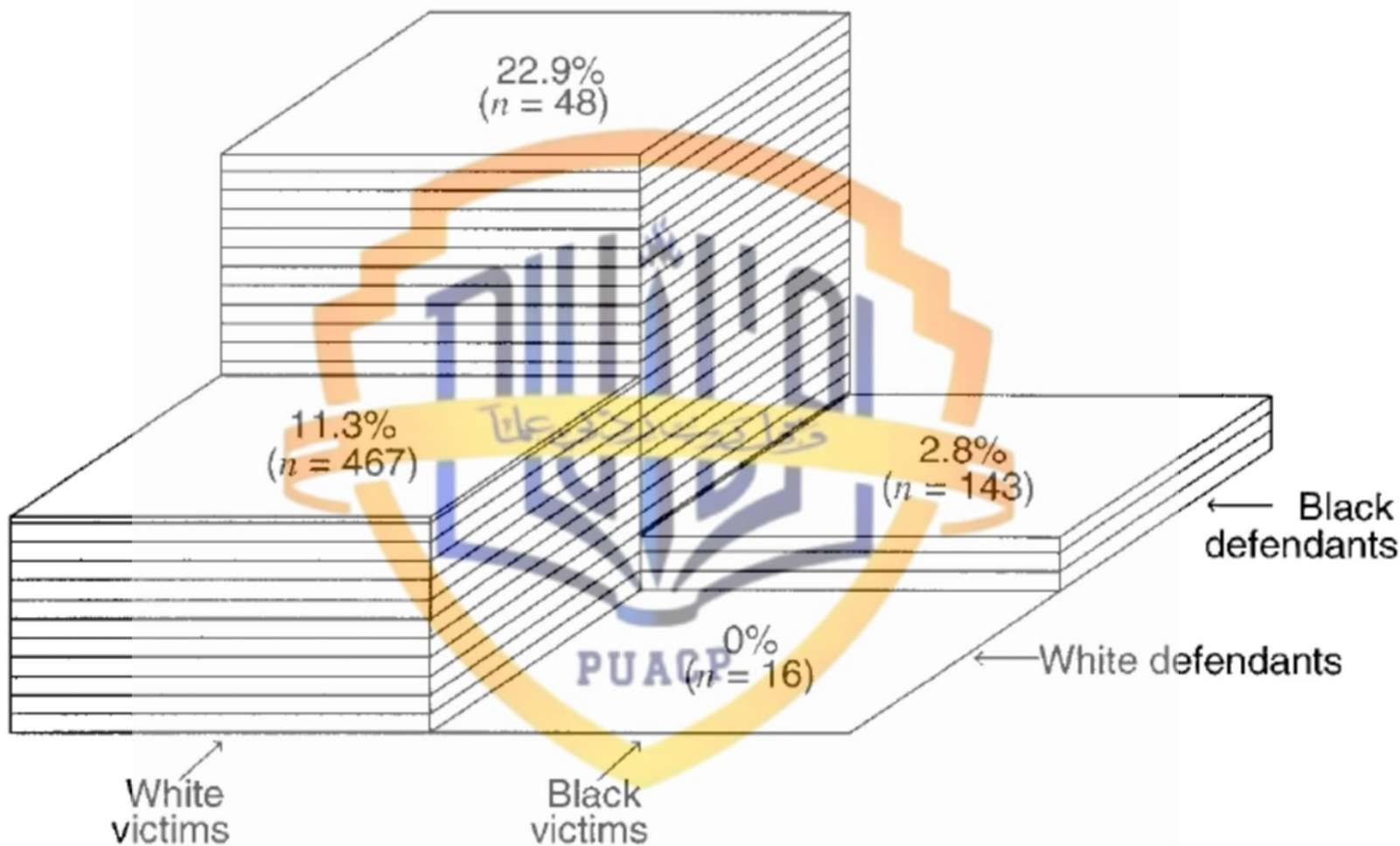
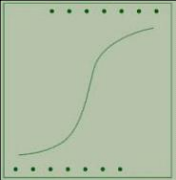
PUACP

2.3.2 Death Penalty Example

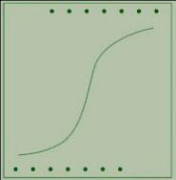
TABLE 2.6 Death Penalty Verdict by Defendant's Race and Victims' Race

Victims' Race	Defendant's Race	Death Penalty		Percent Yes
		Yes	No	
White	White	53	414	11.3
	Black	11	37	22.9
Black	White	0	16	0.0
	Black	4	139	2.8
Total	White	53	430	11.0
	Black	15	176	7.9

It studied effects of racial characteristics on whether persons convicted of homicide received the death penalty.



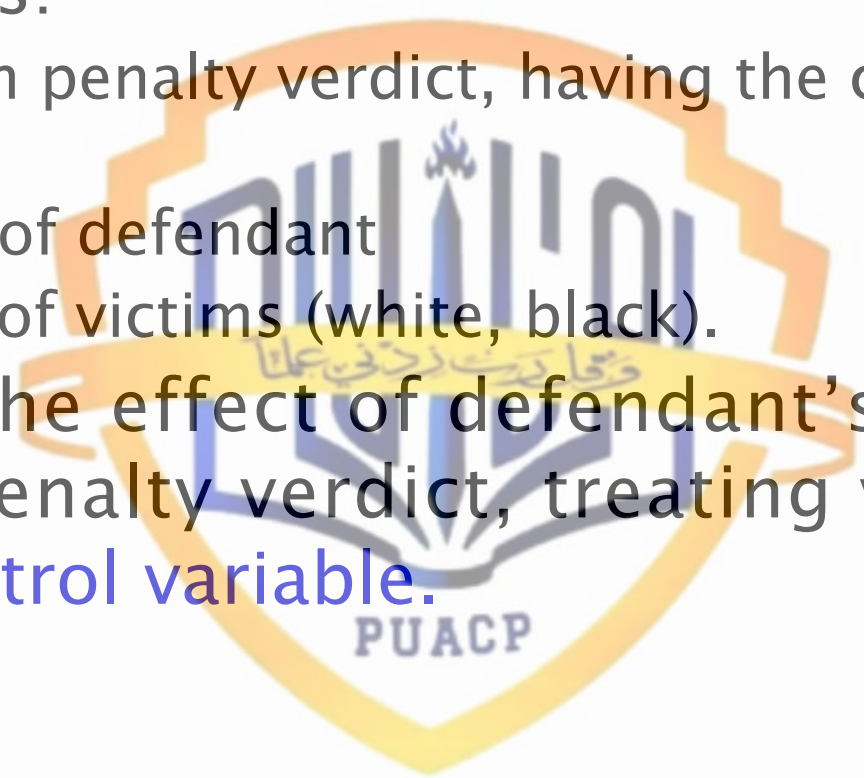
Percent receiving death penalty.

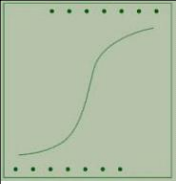


► Variables:

- Y=death penalty verdict, having the categories (yes, no),
- X=race of defendant
- Z=race of victims (white, black).

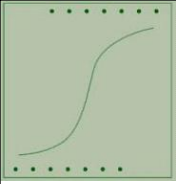
► Study: the effect of defendant's race on the death penalty verdict, treating victims' race as a **control variable**.





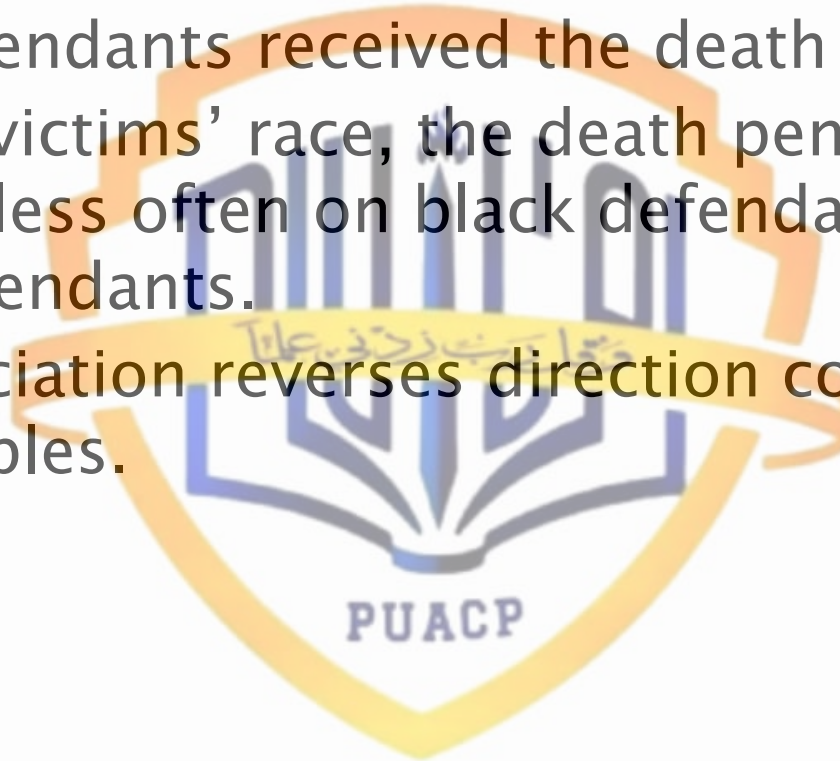
Conditional Y^*X given Z

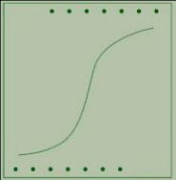
- ▶ When the victims were white, the death penalty was imposed $22.9\% - 11.3\% = 11.6\%$ more often for black defendants than for white defendants.
- ▶ When the victims were black, the death penalty was imposed 2.8% more often for black defendants than for white defendants.
- ▶ Controlling for victims' race by keeping it fixed, the death penalty was imposed more often on black defendants than on white defendants.



Marginal Y^*X

- ▶ Overall, 11.0% of white defendants and 7.9% of black defendants received the death penalty.
- ▶ Ignoring victims' race, the death penalty was imposed less often on black defendants than on white defendants.
- ▶ The association reverses direction compared to the partial tables.





Why does the association change so much when we ignore control variable victims' race?

- ▶ This relates to the nature of the association between victims' race and each of the other variables.
- ▶ First, the association between victims' race and defendant's race is extremely strong.

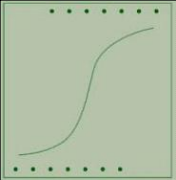
Marginal table Z*X

(I) The marginal table relating these variables has odds ratio $(467 \times 143) / (48 \times 16) = 87.0$

So whites are tending to kill whites

X
PUACP
Z

Frequency	Black	White	Total
Black	143	48	191
White	16	467	483
Total	159	515	674



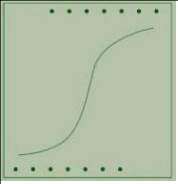
Marginal table $Z*Y$

(II) regardless of defendant's race, the death penalty was much more likely when the victims were white than when the victims were black.

		Y		Z		
Frequency		Black	White		Total	
No	155	451	606			
Yes	4	64	68			
Total	159	515	674			

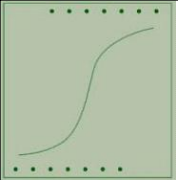
Killing whites is more likely to result in the death penalty.

the marginal association should show a greater tendency than the conditional associations for white defendants to receive the death penalty.



Simpson's paradox (Simpson 1951, Yule 1903)

- ▶ The result that a marginal association can have a **different direction** from each conditional association is called Simpson's paradox
- ▶ It applies to quantitative as well as categorical variables.
- ▶ Statisticians commonly use it to caution against imputing causal effects from an association of X with Y.
- ▶ For instance, when doctors started to observe strong odds ratios between smoking and lung cancer, Statisticians such as R. A. Fisher warned that some variable (e.g., a genetic factor) could exist such that the association would disappear under the relevant control.
- ▶ However, with a very strong XY association, a very strong association must exist between the confounding variable Z and both X and Y in order for the effect to disappear or change under the control.



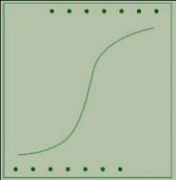
2.3.3 Conditional and Marginal Odds Ratios for $2 \times 2 \times K$ tables

- ▶ for $2 \times 2 \times K$ tables, where K denotes the number of categories of a control variable
- ▶ Let $\{\mu_{ijk}\}$ denote cell expected frequencies for some sampling model, such as binomial, multinomial, or Poisson sampling.
- ▶ Within a fixed category k of Z , the odds ratio

$$\theta_{XY(k)} = \frac{\mu_{11k}\mu_{22k}}{\mu_{12k}\mu_{21k}} \quad \text{or sample OR} \quad \hat{\theta}_{XY(k)} = \frac{n_{11k}n_{22k}}{n_{12k}n_{21k}}$$

describes conditional XY association in partial table k .

- ▶ The odds ratios for the K partial tables are called XY conditional odds ratios.

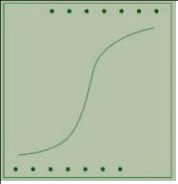


▶ *XY marginal odds ratio* $\theta_{XY} = \frac{\mu_{11+}\mu_{22+}}{\mu_{12+}\mu_{21+}}$

▶ *sample OR* $\hat{\theta}_{XY} = \frac{n_{11+}n_{22+}}{n_{12+}n_{21+}}$

TABLE 2.6 Death Penalty Verdict by Defendant's Race and Victims' Race

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Total	White	53	430	11.0
	Black	15	176	7.9



association between defendant's race and the death penalty

- ▶ Given victims' race is white

$$\hat{\theta}_{XY(1)} = \frac{53 \times 37}{414 \times 11} = 0.43$$

The sample odds for white defendants receiving the death penalty were 43% of the sample odds for black defendants.

- ▶ Given victims' race is black

$$\hat{\theta}_{XY(2)} = \frac{0 \times 139}{16 \times 4} = 0$$

- ▶ Estimation of the marginal odds ratio

$$\hat{\theta}_{XY} = \frac{53 \times 176}{430 \times 15} = 1.45$$

- ▶ The sample odds of the death penalty were 45% higher for white defendants than for black defendants.

Example

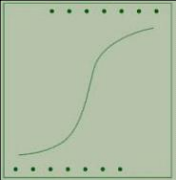
TABLE 2.7 Expected Frequencies Showing That Conditional Independence Does Not Imply Marginal Independence

Clinic	Treatment	Response	
		Success	Failure
1	A	18	12
	B	12	8
2	A	2	8
	B	8	32
Total	A	20	20
	B	20	40

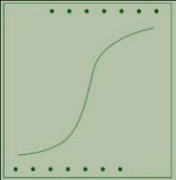
$$\theta_{XY(1)} = \frac{18 \times 8}{12 \times 12} = 1.0, \quad \theta_{XY(2)} = \frac{2 \times 32}{8 \times 8} = 1.0.$$

- ▶ Given the clinic, response and treatment are conditionally independent.
- ▶ Ignore the clinical, response and treatment are not marginally independent.

$$\theta_{XY} = (20 \times 40) / (20 \times 20) = 2.0$$



- ▶ Ignoring the clinic, why are the odds of a success for treatment A twice those for treatment B?
- ▶ The conditional XZ and YZ odds ratios give a clue.
- ▶ The odds ratio between Z and either X or Y, at each fixed category of the other variable, equals 6.0. For instance, the XZ odds ratio at the first category of Y equals $18 \times 8 / 12 \times 2 = 6.0$.
- ▶ The conditional odds (given response) of receiving treatment A at clinic 1 are six times those at clinic 2, and the conditional odds (given treatment) of success at clinic 1 are six times those at clinic 2.
- ▶ Clinic 1 tends to use treatment A more often, and clinic 1 also tends to have more successes.
- ▶ For instance, if patients at clinic 1 tended to be younger and in better health than those at clinic 2, perhaps they had a better success rate regardless of the treatment received.



2.3.5 Homogeneous Association

A $2 \times 2 \times K$ table has *homogeneous XY association* when

$$\theta_{XY(1)} = \theta_{XY(2)} = \cdots = \theta_{XY(K)}.$$

- ▶ Then the effect of X on Y is the same at each category of Z.
- ▶ Conditional independence of X and Y is the special case in which each $\theta_{XY(k)} = 1.0$.
- ▶ Under homogeneous XY association, homogeneity also holds for the other associations. (symmetric)
- ▶ When it occurs, there is said to be **no interaction** between two variables in their effects on the other variable.